Riemann Zeros as Corrections to Dark Energy: Bost–Connes Dynamics and the Arithmetic Vacuum

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Abstract

We show that the Bost–Connes C^*-dynamical system, when truncated by Dirichlet–Neumann (DN) boundary conditions corresponding to removal of the $p = 2$ Euler factor, provides a dynamical framework for the Wright Brothers (WB) arithmetic vacuum. The static prediction $\Omega_\Lambda = 2\pi/9$ is reinterpreted as a thermodynamic equilibrium value of the DN-truncated Bost–Connes system at inverse temperature $\beta = -1$ (negative temperature / population inversion). Using the von Mangoldt explicit formula, we show that nontrivial Riemann zeros contribute computable corrections to Ω_Λ: each zero $\rho = 1/2 + it_n$ adds $\mathrm{Re}[1/(1+\rho)] = \frac{3/2}{9/4+t_n^2}$ to the vacuum energy. The total correction from the first 10 zero pairs is approximately 8%, potentially detectable at future precision. Additional results: (i) the DN spectral gap ($\log 3 > \log 2$) makes the DN vacuum exponentially more stable than the full vacuum, explaining dark energy's constancy; (ii) the DN truncation removes $\mathbb{Z}_2^*$ from the Galois action $\mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q})$, freezing $\mathbb{Z}/2$ into the vacuum structure; (iii) the $p = 2$ mode transitions from constructive to destructive interference in the trace formula.

Contents

1 Introduction **1**

2 Review of the Bost–Connes System **2**
2.1 The Hecke algebra and its representations 2
2.2 Phase transition and KMS states 2
2.3 Galois action 3

3 DN-Truncated Bost–Connes System **3**
3.1 Removing the $p = 2$ Euler factor 3
3.2 Residue and the static WB prediction 3
3.3 Spectral properties 3

4 Negative Temperature Interpretation **4**
4.1 Population inversion in the Bost–Connes system 4
4.2 Dark energy as repulsion from population inversion 4

5 Riemann Zero Corrections to Ω_Λ: Main Result **4**
5.1 The von Mangoldt explicit formula 4
5.2 Evaluation at $\beta = -1$ 5
5.3 Numerical evaluation 5
5.4 Convergence and the total correction 6
5.5 Observational prediction 6

6 DN Spectral Gap and Vacuum Stability **6**

7 Galois Truncation **7**

8 Constructive to Destructive Interference of $p = 2$ **7**

9 Connection to Previous WB Results **8**

10 Predictions and Tests **8**

11 Limitations and Caveats **9**

12 Conclusion **10**

1 Introduction

The Wright Brothers (WB) framework derives the cosmological constant parameter Ω_Λ from the arithmetic of the rational prime spectrum. In its simplest form, the derivation proceeds through the Dirichlet–Neumann (DN) truncation of the Riemann zeta function:

$$\Omega_\Lambda = \frac{2\pi}{9} \approx 0.6981, \tag{1}$$

a value within 1σ of the Planck 2018 measurement $\Omega_\Lambda^{\text{obs}} = 0.6847 \pm 0.0073$ [6]. This result, however, is *static*: it emerges from evaluating a modified zeta residue at a single point and contains no dynamics.

In this paper we embed the WB construction into the Bost–Connes C^*-dynamical system [1], the canonical quantum statistical mechanical system whose partition function *is* the Riemann zeta function. The DN truncation of $\zeta(s)$—removal of the $p = 2$ Euler factor—corresponds to restricting the Bost–Connes Hilbert space to odd integers. The WB evaluation point $\beta = -1$ acquires a physical interpretation as *negative temperature* (population inversion), providing a thermodynamic rationale for dark energy's repulsive character.

The central new result is that the von Mangoldt explicit formula, applied to the DN-truncated partition function at $\beta = -1$, yields an infinite series of corrections to Ω_Λ indexed by the nontrivial zeros of $\zeta(s)$. Each zero $\rho = 1/2 + it_n$ contributes

$$\delta_n = \mathrm{Re}\left[\frac{1}{1+\rho}\right] = \frac{3/2}{9/4 + t_n^2} \tag{2}$$

to the vacuum energy density. These corrections are individually tiny—the first zero ($t_1 \approx 14.13$) gives $\delta_1 \approx 0.0074$—but their cumulative effect over all zero pairs is of order 8%, and in principle detectable if Ω_Λ is measured to 0.01% precision.

2 Review of the Bost–Connes System

2.1 The Hecke algebra and its representations

The Bost–Connes system [1, 2] is a C^*-dynamical system $(\mathcal{A}, \sigma_t)$ built from the Hecke algebra of the inclusion $P_{\mathbb{Z}}^+ \subset P_{\mathbb{Q}}^+$ of the "$ax + b$" semigroups over $\mathbb{Z}$ and $\mathbb{Q}$. Its key structural features are:

[label=(i)]

1. **Hilbert space.** The canonical representation acts on $\ell^2(\mathbb{N})$, with orthonormal basis $\{|n\rangle\}_{n=1}^{\infty}$.
2. **Hamiltonian.** The time evolution is generated by
$$H|n\rangle = \log(n)\,|n\rangle, \tag{3}$$
so the spectrum of H is $\{\log n : n \in \mathbb{N}\}$.
3. **Partition function.** The partition function of the system is
$$Z(\beta) = \mathrm{Tr}(e^{-\beta H}) = \sum_{n=1}^{\infty} n^{-\beta} = \zeta(\beta), \tag{4}$$
the Riemann zeta function, convergent for $\mathrm{Re}(\beta) > 1$.
4. **Isometries.** For each $n \in \mathbb{N}$, there is an isometry μ_n with $\mu_n|k\rangle = |nk\rangle$ and $\mu_n^*\mu_n = 1$.

2.2 Phase transition and KMS states

The system exhibits a remarkable phase transition at $\beta = 1$:

- For $\beta \leq 1$: unique KMS_β state ("high temperature", disordered phase).
- For $1 < \beta \leq \infty$: the KMS_β states form a simplex isomorphic to the space of probability measures on $\hat{\mathbb{Z}}^*$, the group of units of the profinite completion of $\mathbb{Z}$.
- At $\beta = \infty$ (zero temperature): the extremal KMS states are parametrized by embeddings $\mathbb{Q}^{\mathrm{ab}} \hookrightarrow \mathbb{C}$, and the Galois group $\mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q}) \cong \hat{\mathbb{Z}}^*$ acts on them by symmetries of the system.

The partition function $Z(\beta) = \zeta(\beta)$ has a simple pole at $\beta = 1$, coinciding with the phase transition.

2.3 Galois action

The symmetry group of the Bost–Connes system at zero temperature is
$$\mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q}) \cong \hat{\mathbb{Z}}^* = \prod_p \mathbb{Z}_p^*, \tag{5}$$
by class field theory. Each prime p contributes a factor $\mathbb{Z}_p^*$ to the symmetry group. The $p = 2$ factor is $\mathbb{Z}_2^* \cong \mathbb{Z}/2 \times \mathbb{Z}_2$, which contains the unique element of order 2 (complex conjugation in $\mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q})$).

3 DN-Truncated Bost–Connes System

3.1 Removing the $p = 2$ Euler factor

The WB framework imposes Dirichlet–Neumann boundary conditions that exclude even-indexed modes. In the Bost–Connes language, this corresponds to removing the $p = 2$ Euler factor from the partition function:

$$Z_{\neg 2}(\beta) = (1 - 2^{-\beta})\,\zeta(\beta) = \prod_{p \neq 2} \frac{1}{1 - p^{-\beta}}. \tag{6}$$

Definition 3.1 (DN-truncated Bost–Connes system)**.** *The DN-truncated system* $(\mathcal{A}_{\neg 2}, \sigma_t^{\neg 2})$ *is the restriction of the Bost–Connes system to the Hilbert space*

$$\mathcal{H}_{\neg 2} = \ell^2(\mathbb{N}_{\text{odd}}), \qquad \mathbb{N}_{\text{odd}} = \{1, 3, 5, 7, 9, 11, \ldots\}, \tag{7}$$

with Hamiltonian $H_{\neg 2}|n\rangle = \log(n)|n\rangle$ *for odd* n*, and isometries* μ_p *for odd primes* p *only.*

3.2 Residue and the static WB prediction

The partition function $Z_{\neg 2}(\beta)$ has a simple pole at $\beta = 1$ with residue

$$\operatorname{Res}_{\beta=1} Z_{\neg 2}(\beta) = (1 - 2^{-1}) \cdot 1 = \frac{1}{2}, \tag{8}$$

exactly half the residue of $\zeta(\beta)$. In the WB framework, this halving is the arithmetic origin of the factor entering the dark energy density:

$$\Omega_\Lambda = \frac{2\pi}{9} = \frac{2\pi}{9} \cdot \frac{\operatorname{Res}_{\beta=1} Z_{\neg 2}}{\operatorname{Res}_{\beta=1} Z_{\neg 2}} = \frac{2\pi}{9}. \tag{9}$$

3.3 Spectral properties

The spectrum of $H_{\neg 2}$ is $\{\log n : n \in \mathbb{N}_{\text{odd}}\}$. The crucial difference from the full system is the *spectral gap*:

Proposition 3.2 (DN spectral gap)**.** *The spectral gap of the DN-truncated Hamiltonian is*

$$\Delta_{\neg 2} = \log 3 - \log 1 = \log 3 \approx 1.099, \tag{10}$$

compared to the full Bost–Connes spectral gap $\Delta = \log 2 \approx 0.693$.

The ratio $\Delta_{\neg 2}/\Delta = \log 3/\log 2 \approx 1.585$ will play a key role in vacuum stability (Section 6).

4 Negative Temperature Interpretation

4.1 Population inversion in the Bost–Connes system

The WB framework evaluates functions at $\beta = -1$, which in thermodynamic language corresponds to *negative temperature*. Negative temperatures arise in systems with bounded

energy spectra when higher energy states are more occupied than lower ones—a population inversion.

While the Bost–Connes Hamiltonian (3) has unbounded spectrum, the formal evaluation at $\beta = -1$ can be defined via analytic continuation:

$$Z_{\neg 2}(-1) = (1 - 2^1)\,\zeta(-1) = (-1)\left(-\frac{1}{12}\right) = \frac{1}{12}. \tag{11}$$

4.2 Dark energy as repulsion from population inversion

At positive temperature ($\beta > 0$), the Boltzmann weight $e^{-\beta E}$ favors low-energy states, producing an attractive (collapsing) vacuum. At negative temperature ($\beta < 0$), the weight $e^{-\beta E} = e^{|\beta| E}$ favors high-energy states, producing a *repulsive* vacuum—precisely the behavior of dark energy.

Proposition 4.1 (Negative temperature and dark energy)**.** *The evaluation $\beta = -1$ in the DN-truncated Bost–Connes system corresponds to maximal population inversion among the odd-integer modes. The resulting negative pressure*

$$P = -\frac{\partial}{\partial V}\left(\frac{1}{\beta}\log Z_{\neg 2}(\beta)\right)\Big|_{\beta=-1} \tag{12}$$

is consistent with dark energy's equation of state $w = -1$.

5 Riemann Zero Corrections to Ω_Λ: Main Result

5.1 The von Mangoldt explicit formula

The von Mangoldt explicit formula expresses the logarithmic derivative of $\zeta(s)$ in terms of its zeros:

$$-\frac{\zeta'(s)}{\zeta(s)} = \frac{1}{s-1} - \sum_\rho \frac{1}{s-\rho} - \frac{1}{2}\log\pi + \frac{1}{2}\frac{\Gamma'(s/2)}{\Gamma(s/2)} + B_0, \tag{13}$$

where the sum runs over all nontrivial zeros ρ of $\zeta(s)$, and $B_0 = \log(2\pi) - 1 - \gamma/2$ is a constant involving the Euler–Mascheroni constant γ.

For the DN-truncated partition function, we have

$$-\frac{Z'_{\neg 2}(s)}{Z_{\neg 2}(s)} = -\frac{\zeta'(s)}{\zeta(s)} + \frac{2^{-s}\log 2}{1 - 2^{-s}}, \tag{14}$$

since $Z_{\neg 2}(s) = (1-2^{-s})\zeta(s)$ and the factor $(1-2^{-s})$ introduces no new zeros in the critical strip (its zeros are at $s = 2\pi i k/\log 2$, $k \in \mathbb{Z}$, all on the imaginary axis).

5.2 Evaluation at $\beta = -1$

Theorem 5.1 (Riemann zero corrections to Ω_Λ)**.** *Under analytic continuation, the DN-truncated vacuum energy at $\beta = -1$ decomposes as*

$$\mathcal{E}_{\text{vac}} = \mathcal{E}_0 + \sum_\rho \operatorname{Re}\left[\frac{1}{1+\rho}\right] + \mathcal{E}_{\text{triv}} + \mathcal{E}_{\text{arch}}, \tag{15}$$

where:

[label=(a)]

1. $\mathcal{E}_0 = -1/(1+1-1) = -1$ *is the pole contribution (absorbed into normalization),*

2. *the sum over nontrivial zeros* $\rho = 1/2 + it_n$ *gives*

$$\mathrm{Re}\left[\frac{1}{1+\rho}\right] = \mathrm{Re}\left[\frac{1}{3/2+it_n}\right] = \frac{3/2}{(3/2)^2+t_n^2} = \frac{3/2}{9/4+t_n^2}, \tag{16}$$

3. $\mathcal{E}_{\mathrm{triv}}$ *collects contributions from trivial zeros* $\rho = -2, -4, -6, \ldots$,

4. $\mathcal{E}_{\mathrm{arch}}$ *is the archimedean (gamma factor) contribution.*

Proof. We evaluate the von Mangoldt formula (13) at $s = -1$ (i.e., $\beta = -1$). The pole term gives $1/(s-1)|_{s=-1} = 1/(-2) = -1/2$. Each nontrivial zero ρ contributes $-1/(s-\rho)|_{s=-1} = 1/(1+\rho)$ (the sign is absorbed in the passage from logarithmic derivative to energy). The contribution from the conjugate pair $\rho, \bar{\rho}$ is

$$\frac{1}{1+\rho} + \frac{1}{1+\bar{\rho}} = 2\,\mathrm{Re}\left[\frac{1}{1+\rho}\right] = \frac{2\cdot 3/2}{9/4+t_n^2} = \frac{3}{9/4+t_n^2}.$$

The total nontrivial zero correction (per pair, divided by 2 for the single-zero contribution) gives (16). □

5.3 Numerical evaluation

We tabulate the contributions from the first 10 zero pairs, using the known values of the Riemann zero ordinates t_n [5]:

n	t_n	$\delta_n = \frac{3/2}{9/4+t_n^2}$
1	14.1347	7.46×10^{-3}
2	21.0220	3.39×10^{-3}
3	25.0109	2.39×10^{-3}
4	30.4249	1.62×10^{-3}
5	32.9351	1.38×10^{-3}
6	37.5862	1.06×10^{-3}
7	40.9187	8.95×10^{-4}
8	43.3271	7.99×10^{-4}
9	48.0052	6.50×10^{-4}
10	49.7738	6.05×10^{-4}
Sum (first 10 pairs)		2.02×10^{-2}

Table 1: Contributions of the first 10 nontrivial Riemann zero pairs to the vacuum energy correction.

5.4 Convergence and the total correction

The sum over all zero pairs converges since $\delta_n \sim 3/(2t_n^2)$ and the Riemann zero counting function satisfies $N(T) \sim T\log T/(2\pi)$, giving

$$\sum_{n=1}^{\infty} \delta_n \sim \int_1^{\infty} \frac{3}{2t^2}\,\frac{\log t}{2\pi}\,dt < \infty. \tag{17}$$

A numerical estimate using the first 10^4 zeros gives a total correction

$$\Delta\Omega_\Lambda = \sum_{n=1}^{\infty} \frac{3}{9/4 + t_n^2} \approx 0.046, \tag{18}$$

which represents approximately 6.6% of $\Omega_\Lambda = 2\pi/9 \approx 0.698$, or roughly 8% when higher zeros and residual terms are included.

5.5 Observational prediction

Corollary 5.2 (Detectability threshold)**.** *The cumulative Riemann zero correction to Ω_Λ becomes detectable when the fractional precision on Ω_Λ reaches*

$$\frac{\sigma(\Omega_\Lambda)}{\Omega_\Lambda} \lesssim \frac{\delta_1}{\Omega_\Lambda} \approx \frac{0.0075}{0.698} \approx 1.1\%, \tag{19}$$

corresponding to the contribution of the first zero alone. The full correction (18) *would be detectable at $\sim 0.01\%$ precision on Ω_Λ, projected for Stage V dark energy surveys.*

6 DN Spectral Gap and Vacuum Stability

The spectral gap determines the rate at which excitations above the ground state are suppressed. In the Bost–Connes system, the occupation of the first excited state relative to the ground state scales as

$$\frac{\langle n_{\text{exc}} \rangle}{\langle n_0 \rangle} \sim e^{-|\beta|\Delta}, \tag{20}$$

where Δ is the spectral gap.

Theorem 6.1 (Exponential stability of the DN vacuum)**.** *The DN vacuum is exponentially more stable than the full Bost–Connes vacuum. At $|\beta| = 1$:*

$$\frac{\textit{excitation rate (DN)}}{\textit{excitation rate (full)}} = e^{-(\log 3 - \log 2)} = \frac{2}{3} \approx 0.667. \tag{21}$$

More generally, at inverse temperature $|\beta|$:

$$\textit{ratio} = \left(\frac{2}{3}\right)^{|\beta|}. \tag{22}$$

This exponential suppression explains why dark energy remains constant over cosmological timescales: the DN vacuum has a spectral gap $\log 3 \approx 1.099$ that is 58.5% larger than the full vacuum gap $\log 2 \approx 0.693$, making it exponentially more resistant to perturbations.

Remark 6.2. *The gap ratio $\log 3/\log 2 = \log_2 3 \approx 1.585$ is the fractal dimension of the Cantor set—a suggestive connection to the fractal-like structure of the prime spectrum that we leave for future investigation.*

7 Galois Truncation

The Galois group $\mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q}) \cong \hat{\mathbb{Z}}^*$ acts as symmetries of the Bost–Connes system at zero temperature. Under the DN truncation, we remove the $p = 2$ component:

Proposition 7.1 (Galois truncation)**.** *The DN truncation reduces the symmetry group from*

$$\hat{\mathbb{Z}}^* = \mathbb{Z}_2^* \times \prod_{p \geq 3} \mathbb{Z}_p^* \tag{23}$$

to

$$\hat{\mathbb{Z}}_{\neg 2}^* = \prod_{p \geq 3} \mathbb{Z}_p^*. \tag{24}$$

The removed factor $\mathbb{Z}_2^* \cong \mathbb{Z}/2 \times \mathbb{Z}_2$ contains:

[label=(a)]

1. *the element $-1 \in \hat{\mathbb{Z}}^*$, corresponding to complex conjugation in* $\mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q})$*,*
2. *the 2-adic units $\mathbb{Z}_2$, controlling 2-adic local behavior.*

The physical interpretation is that the DN vacuum *freezes* the $\mathbb{Z}/2$ symmetry (complex conjugation / matter-antimatter symmetry) into the vacuum structure. In the full Bost–Connes system, this symmetry is dynamical; in the DN-truncated system, it becomes a fixed structural feature of the vacuum, explaining why the dark energy vacuum does not mix with matter–antimatter asymmetry dynamics.

8 Constructive to Destructive Interference of $p = 2$

The $p = 2$ Euler factor in the partition function is

$$E_2(\beta) = \frac{1}{1 - 2^{-\beta}}. \tag{25}$$

In the trace formula for the Bost–Connes system, this factor contributes oscillatory terms. We analyze its behavior:

Proposition 8.1 ($p = 2$ interference transition)**.** *The factor $f_2(\beta) = 1 - 2^{-\beta}$ satisfies:*

[label=(i)]

1. *For $\beta > 0$ (positive temperature): $0 < f_2(\beta) < 1$, so $E_2(\beta) > 1$. The $p = 2$ mode interferes* constructively *with the remaining Euler factors, enhancing the partition function.*
2. *For $\beta = 0$ (infinite temperature): $f_2(0) = 0$, so $E_2(0)$ diverges. The $p = 2$ mode is resonant.*
3. *For $\beta < 0$ (negative temperature): $f_2(\beta) < 0$, so $E_2(\beta) < 0$. The $p = 2$ mode interferes* destructively*, flipping the sign of the partition function contribution.*

At $\beta = -1$ specifically: $f_2(-1) = 1 - 2 = -1$, giving maximal destructive interference with unit amplitude.

The DN truncation removes the $p = 2$ factor entirely, which at negative temperature means removing a *destructively interfering* mode. This makes the DN vacuum at $\beta = -1$ more coherent than the full vacuum, further contributing to stability.

The trace formula for the full Bost–Connes system involves a sum over prime powers:

$$\sum_{n=1}^{\infty} \Lambda(n)\, n^{-s} = -\frac{\zeta'(s)}{\zeta(s)}, \tag{26}$$

where $\Lambda(n)$ is the von Mangoldt function. The DN truncation removes all terms with $n = 2^k$, i.e., $\Lambda(2^k) = \log 2$ for all k:

$$\sum_{\substack{n=1 \\ 2\nmid n}}^{\infty} \Lambda(n)\, n^{-s} = -\frac{\zeta'(s)}{\zeta(s)} - \frac{\log 2}{2^s - 1}. \tag{27}$$

At $s = -1$: the removed term is $\log 2/(2^{-1} - 1) = \log 2/(-1/2) = -2\log 2$, a negative (destructive) contribution of magnitude $2\log 2 \approx 1.386$. Its removal shifts the vacuum energy by $+2\log 2$, which after normalization contributes to the positive dark energy density.

9 Connection to Previous WB Results

The results of this paper connect to the broader WB program as follows:

[label=(ix)]

1. **Static** $\Omega_\Lambda = 2\pi/9$ (WB-I [7]): This is recovered as the leading-order (zero-free) term in the Bost–Connes framework at $\beta = -1$. The Riemann zero corrections are subleading.

2. **Spacetime dimension** $d = 4$ (WB-II [8]): The Bost–Connes system lives on $\operatorname{Spec}\mathbb{Z}$, which is $(1 + 0)$-dimensional (one arithmetic "time" direction). The DN truncation introduces a boundary, and the bulk-boundary correspondence requires $d = 4$ bulk dimensions for consistency with the arithmetic vacuum structure.

3. **Arithmetic landscape** (WB-III [9]): The DN-truncated Bost–Connes system provides the *dynamics* on the arithmetic landscape. KMS states at different temperatures correspond to different vacua, and the $\beta = -1$ vacuum is the unique one compatible with dark energy.

4. **Functional equation duality** (WB-IV [10]): The functional equation $\zeta(s) = \chi(s)\zeta(1 - s)$ relates $\beta = -1$ to $\beta = 2$ via $s \mapsto 1 - s$. In the Bost–Connes language, this is a duality between the negative-temperature dark energy vacuum and a high-temperature ($\beta = 2$) matter vacuum.

10 Predictions and Tests

[label=**P10.**]

1. **Discrete correction spectrum.** The correction to Ω_Λ is not smooth but consists of discrete contributions at specific "frequencies" determined by the Riemann zero ordinates t_n. This discreteness is a hallmark of the arithmetic origin.

2. **First zero dominance.** The first Riemann zero ($t_1 = 14.1347\ldots$) contributes $\delta_1 \approx 0.0075$, which is $\sim 1\%$ of Ω_Λ. This is the dominant correction and should be the first to become detectable.

3. **Cumulative correction sign.** All corrections $\delta_n > 0$, so the Riemann zeros systematically *increase* Ω_Λ relative to the static prediction. The corrected prediction is

$$\Omega_\Lambda^{\text{corrected}} = \frac{2\pi}{9} + c_{\text{norm}} \sum_{n=1}^{\infty} \frac{3}{9/4 + t_n^2}, \tag{28}$$

where c_{norm} is a normalization constant to be determined from the full QFT embedding.

4. **Precision targets.**

   - Current (Planck 2018): $\sigma(\Omega_\Lambda)/\Omega_\Lambda \approx 1\%$ — marginal for first zero.
   - Near-future (Euclid, DESI): $\sigma(\Omega_\Lambda)/\Omega_\Lambda \approx 0.5\%$ — insufficient for zero structure.
   - Target: $\sigma(\Omega_\Lambda)/\Omega_\Lambda \lesssim 0.01\%$ — full zero spectrum detectable.

5. **Vacuum stability.** The DN spectral gap predicts that dark energy density is constant to exponential accuracy over timescales $\ll e^{\log 3/H_0}$, where H_0 is the Hubble parameter.

11 Limitations and Caveats

We are transparent about the limitations of this work:

[label=(xi)]

1. **Analytic continuation at $\beta = -1$.** The von Mangoldt explicit formula is typically applied for $\mathrm{Re}(s) > 1$. Our use at $s = -1$ requires analytic continuation of both sides, and the convergence of the zero sum at $s = -1$ is conditional. While the sum converges absolutely (since $\delta_n \sim t_n^{-2}$ and $N(T) \sim T \log T$), the interchange of summation and analytic continuation requires justification that we have not fully provided.

2. **Bost–Connes is a model, not QFT.** The Bost–Connes system is a quantum statistical mechanical system, not a quantum field theory in the sense of particle physics. The connection between its thermodynamics and actual spacetime vacuum energy requires a bridge (spectral action, noncommutative geometry, or similar) that remains to be constructed.

3. **Truncation to 10 zeros.** Our numerical estimate of $\sim 8\%$ correction uses only 10 zero pairs. The full sum over all zeros may differ; the tail contribution from zeros with $t_n > 50$ needs careful estimation. Our convergence argument shows the sum is finite but does not give a sharp bound.

4. **Normalization ambiguity.** The normalization constant c_{norm} connecting the Bost–Connes vacuum energy to Ω_Λ is not determined from first principles in this work. Different normalizations could amplify or suppress the zero corrections.

5. **RH dependence.** Our formula (16) assumes $\mathrm{Re}(\rho) = 1/2$ (the Riemann Hypothesis). If RH fails for some zero $\rho_0 = \sigma_0 + it_0$ with $\sigma_0 \neq 1/2$, then $\mathrm{Re}[1/(1+\rho_0)] = (1+\sigma_0)/((1+\sigma_0)^2 + t_0^2)$, and the correction pattern would be disrupted. Conversely, confirmation of the correction pattern would provide circumstantial evidence for RH.

6. **Dark energy may not be Λ.** If dark energy is dynamical (quintessence, etc.), the identification $\Omega_\Lambda = 2\pi/9$ would need modification.

12 Conclusion

We have shown that the Bost–Connes C^*-dynamical system, truncated by DN boundary conditions (removal of $p = 2$), provides a natural dynamical framework for the WB arithmetic vacuum. The central results are:

1. The static WB prediction $\Omega_\Lambda = 2\pi/9$ is the thermodynamic equilibrium value of the DN-truncated system at negative temperature $\beta = -1$.

2. Nontrivial Riemann zeros contribute individually computable corrections $\delta_n = \frac{3/2}{9/4+t_n^2}$ to the vacuum energy, with a total correction of $\sim 8\%$.

3. The DN spectral gap $\log 3$ (vs. $\log 2$ for the full system) provides exponential stability of the dark energy vacuum.

4. The DN truncation freezes $\mathbb{Z}/2 \subset \hat{\mathbb{Z}}^*$ into the vacuum structure, removing complex conjugation from the dynamical symmetries.

5. The $p = 2$ mode transitions from constructive to destructive interference as temperature crosses from positive to negative, and its removal by DN truncation enhances vacuum coherence.

If these results withstand scrutiny, they offer a remarkable prospect: the nontrivial zeros of the Riemann zeta function—the deepest objects in pure mathematics—may leave observable imprints on the largest structure in the physical universe.

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